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Review article

Virtual reality and augmented reality— emerging screening and diagnostic techniques in ophthalmology: A systematic review

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A B S T R A C T

In health care, virtual reality (VR) and augmented reality (AR) have been applied extensively for many purposes. Similar to other technologies such as telemedicine and artificial intelligence, VR and AR may improve clinical diagnosis and screening services in ophthalmology by alleviating current problems, including workforce shortage, diagnostic error, and underdiagnosis. In the past decade a number of studies and products have used VR and AR concepts to build clinical tests for ophthalmology, but comprehensive reviews on these studies are limited. Therefore, we conducted a systematic review on the use of VR and AR as a diagnostic and screening tool in ophthalmology. We identified 26 studies that implemented a variety of VR and AR tests on different conditions, including VR cover tests for binocular vision disorder, VR perimetry for glaucoma, and AR slit lamp biomicroscopy for retinal diseases. In general, while VR and AR tools can become standardized, automated, and cost-effective tests with good user experience, several weaknesses, including unsatisfactory accuracy, weak validation, and hardware limitations, have prevented these VR and AR tools from having wider clinical application. Also, a comparison between VR and AR is made to explain why studies have predominantly used VR rather than AR.

1. Introduction

In health care, virtual reality (VR) and augmented reality (AR) have been applied for various purposes, including post-stroke cognitive rehabilitation, surgical training, laparoscopic and robotic surgeries.[2,36] By 2023, the market for VR and AR technologies in health care application will potentially reach nearly US$20 billion.[A] VR technologies create a simulated world, providing a seemingly real and possibly interactive experience within the virtual space.[13,22] In contrast, AR tech-

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